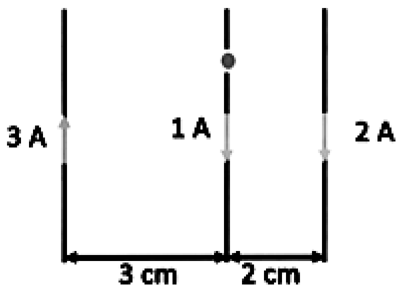
JEE-Main-28-01-2026 (Memory Based)
[MORNING SHIFT]
Physics

Question: There are three long parallel wires in a plane as shown. Find force on 15 cm of length of middle wire.



Options:
(a) 5 μN
(b) 7 μN
(c) 6 μN
(d) 1 μN
Answer: (c)
Solution:

$$f_{net} = 15 \times \cancel{10^{-2}} \times \frac{2}{\cancel{10^{-2}}} \times \left(\frac{\mu_0}{2\pi}\right)$$

$$= 2 \times 10^{-7} \times 15 \times 2$$

$$= 6 \mu N$$

$$f_1 = \frac{\mu_0 I_1 I_2}{2\pi R} = \frac{\mu_0 2}{2\pi (10^{-2})}$$

$$f_2 = \frac{\mu_0 I_2 I_3}{2\pi R'} = \frac{\mu_0}{2\pi}(10^{-2})$$

Question: Equation of an EMW in a medium is given by E = 2 sin(2 × 10^{15}t - 10^{7}x). Find refractive index of the medium.
Options:
(a) 3/2
(b) 2
(c) 5/3
(d) 4/3
Answer: (a)
Solution:

$$C_m = \frac{\omega}{K} = 2 \times 10^8 \, m/s$$

$$\mu = \frac{3 \times 10^8}{2 \times 10^8} = \frac{3}{2}$$

Question: For a circular coil of radius R, center is B_0 = 16 μT. What will be the magnetic field on axis at a distance x = √3 R from center?

Options:

(a) 1/4 μT

(b) 1/2 μT

(c) 4 μT

(d) 2 μT

Answer: (d)

Solution:

$$B(x) = \frac{\mu_0 I}{2} \frac{R^2}{(x^2+R^2)^{3/2}}$$

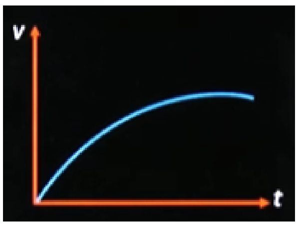
$$\frac{\mu_0 I}{2R} = 16 \, \mu T$$

$$B(x) = \frac{\mu_0 I}{2R} \frac{R^3}{(R^2+x^2)^{3/2}} = 16 \frac{R^3}{(4R^2)^{3/2}} = \frac{16}{8} \frac{\cancel{R^3}}{\cancel{R^3}} = 2 \mu T$$

Question: An object is being dropped from height h above the ground. A part from force of gravity additional drag force, F = -kv acts on the object. Find the graph of v versus t.

Options:

(a) 

(b) 

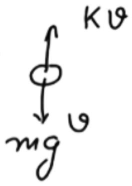
(c) 

(d) 

Answer: (b)

Solution:

$f_{net} = ma = -Kv + mg$



$t = t$

$$\frac{dv}{dt} = g - \frac{kv}{m}$$

$$\int_0^v \frac{dv}{g - \frac{kv}{m}} = \int_{t=0}^{t} dt$$

$$-\frac{m}{K}\left[\ln\left|g - \frac{kv}{m}\right|\right]_0^v = t$$

Question: Electric current in a circuit is given by $i = i_0\left(\frac{t}{T}\right)$. **Find rms current for period t = 0 to t = T.**

Options:

(a) $\frac{i_0}{\sqrt{5}}$

(b) $\frac{i_0}{\sqrt{2}}$

(c) $\frac{i_0}{2}$

(d) $\frac{i_0}{\sqrt{3}}$

Answer: (d)

Solution:

$$i_{rms}^2 = \frac{\int i^2 dt}{\int dt} \Rightarrow i_{rms}^2 = \frac{\frac{i_0^2}{T^2}\int_0^T t^2 dt}{\int_0^T dt}$$

$$i_{rms} = \frac{i_0}{\sqrt{3}}$$

Question: Position of a particle is given by x = A sin(ωt) potential energy is minimum at $t = \frac{T}{2\beta}$, where T is time period. Find minimum value of positive β.

Options:

(a) 1/2

(b) 1

(c) 1/3

(d) 1/6

Answer: (b)

Solution:

$$PE \propto x^2 \propto \underline{\underline{\sin^2 \omega t}}$$

min.

$$\omega t = 0, \pi$$

$$t = \frac{\pi}{2\pi}T = \frac{T}{2}$$

Question: A tap is at a height of 5 m from ground. Water drops are falling from it at regular interval. When 1st drop hits the ground 6th droplet is just about to fall. Find the height of 4th droplet from ground when 1st droplet hits the ground.

Options:

(a) 4.2 m

(b) 3.2 m

(c) 4 m

(d) 3 m

Answer: (a)

Solution:

$$5 = \frac{1}{2} g (5t_0)^2 \Rightarrow g t_0^2 = \frac{2}{5}$$

6^{th}, S_4, 4^{th}

$$S_4 = \frac{1}{2}(2\, t_0)^2 g$$

$$= 2 g t_0^2 = 4/5$$

$$\Delta S_{14} = 5 - 4/5 = \frac{21}{5} = 4.2\,m$$

Question: If 10 kg of ice at -10℃ is mixed with 100 kg of water at 25℃, then resultant temperature in equilibrium for mixture shall be (S_i = 1/2 cal/gam-℃, S_w = 1 cal/gm-℃, L_f = 80 cal/gm)

Options:

(a) 0 ℃

(b) 15 ℃

(c) 12.5 ℃

(d) 5 ℃

Answer: (b)

Solution:

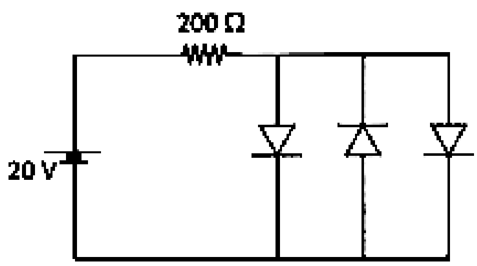
at 0°

Heat absorbtion 10kg ice	Heat Released 100kgW
$m_i s_i 10 + L_f \times 10$	$m_w s_w \Delta T$
$10 \times \frac{1}{2} \times 10 + 80 \times 10$	$100 \times 1 \times 25$
850 kCal	2500 kCal
	$\boxed{1650\ \text{kCal}}$

$$\underline{110} \times 1 \times (T - 0) = 1650\ \text{kcal}$$

$$T = 15°C$$

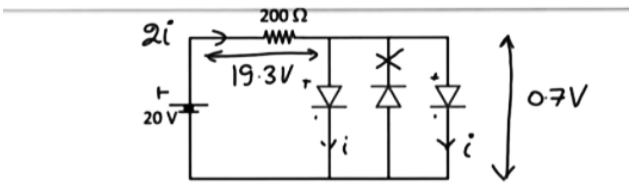
Question: The threshold voltage for the diodes is 0.7 volt. Then current through diodes (from left to right) in given circuit is



Options:

(a) Zero, Zero, Zero

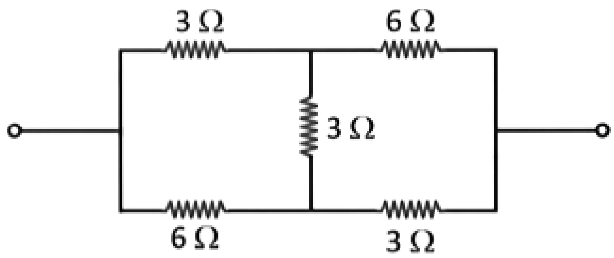
(b) 32.23 mA, 32.23 mA, 32.23 mA
(c) 48.25 mA, Zero, 48.25 mA
(d) 50 mA, Zero, 50 mA
Answer: (c)
Solution:



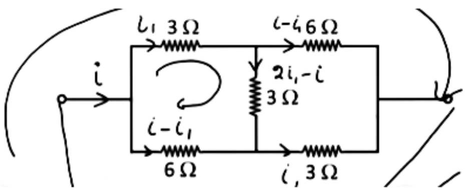
$$2i = \frac{19.3}{200} = 48.25\,mA$$

$$i = 48.25$$

Question: Find equivalent resistance of the given circuit.



Options:
(a) 6.4 Ω
(b) 4.2 Ω
(c) 7 Ω
(d) 5 Ω
Answer: (b)
Solution:



$$-3i_1 - 3(2i_1 - i) + 6(i - i_1) = 0$$

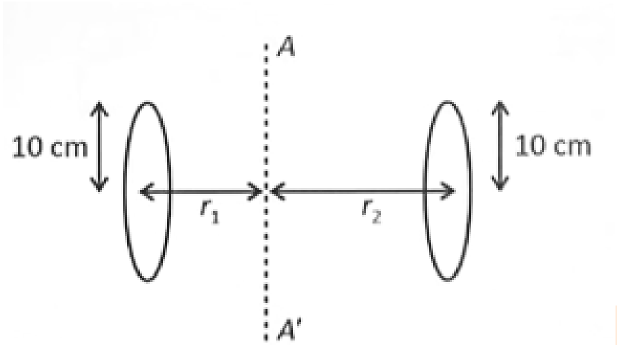
$$\frac{3}{5}i = i_1$$

$$V = 3i_1 + 6(i - i_1)$$

$$V = \frac{9}{5}i + 6 \times \frac{2}{5}i$$

$$R_{eq} = \frac{V}{i} \longrightarrow R_{eq} = \frac{V}{i} = \frac{21}{5} = 4.2\,\Omega$$

Question: For the given situation shown in figure two disks each of mass m = 600 grams are rotating about a fixed axis AA'. Radius of each disk is r_0 = 10 cm and they are at distance r_1 = 10 cm and r_2 = 20 cm from the axis AA'. Torque acting about the axis is 45 × 10^2 dyne-cm. Find angular acceleration in rad/sec^2.



Options:

(a) $\frac{170}{11} rad/\sec^2$

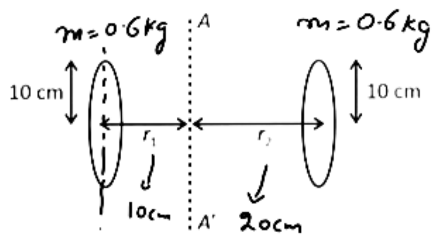
(b) $\frac{140}{9} rad/\sec^2$

(c) $\frac{150}{11} rad/\sec^2$

(d) $\frac{160}{9} rad/\sec^2$

Answer: (c)

Solution:



$\tau = I\alpha$

$\downarrow$

$45 \times 10^2 \times 10^{-5-2}$ Nm

$0.45 = 45 \times 10^{-2}$

$\alpha = \frac{450}{33} = \frac{150}{11}$ rad/sec^2

$I_{AA'} = \frac{mr_0^2}{4} \times 2 + mr_1^2 + mr_2^2$

$m\left(\frac{1}{2 \times 100} + \frac{1}{100} + \frac{4}{100}\right)$

$I_{AA'} = \frac{33}{10^3} = 33 \times 10^{-3}$

Question: A bi-convex lens of refractive index 1.5 and planoconcave lens of refractive index 1.7 have same magnitude of power. If 2nd radius of curvature of convex lens is

equal to radius of curvature of planoconcave less. Find ratio of 1st radius of curvature to 2nd radius of curvature of bi-convex lens.

Options:

(a) 3/2

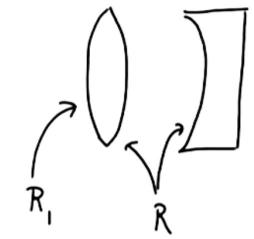
(b) 5/2

(c) 4

(d) 3/4

Answer: (b)

Solution:



$$(1.5-1)\left(\frac{1}{R_1}+\frac{1}{R}\right) = \left|(1.7-1)\left(-\frac{1}{R}\right)\right|$$

$$\frac{1}{2R_1} + \frac{1}{2R} = \frac{7}{10R}$$

$$\frac{1}{2R_1} = \frac{7-5}{10R}$$

$$\frac{R_1}{R} = \frac{5}{2}$$

Question: Find the ratio of de-Broglie wavelength associated with deuteron with kinetic energy of K and α-particle with kinetic energy of 2K.

Options:

(a) 2 : 1

(b) 2√2 : 1

(c) 1 : √2

(d) √2 : 1

Answer: (a)

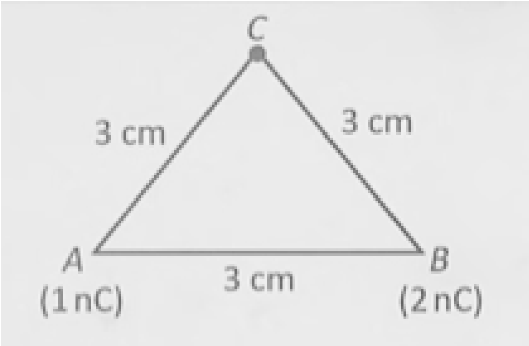
Solution:

$$\lambda = \frac{h}{\sqrt{2m_e K}} \Rightarrow \lambda \propto \frac{1}{\sqrt{mK}}$$

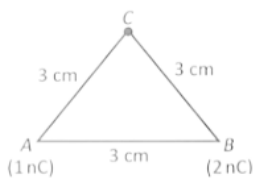
$$\frac{\lambda_d}{\lambda_\alpha} = \frac{\sqrt{4(2K)}}{\sqrt{2K}}$$

$$= 2$$

Question: Find the work done by external agent in moving a 3 nC charge from a large separation to point C.



Options:
(a) 8.1 μJ
(b) 12 μJ
(c) 2.7 μJ
(d) 9 μJ
Answer: (c)
Solution:



$W = U_f - U_i \leftarrow \infty$

$W = qV_c = 3n \times V_c$

$V_c = \frac{K}{3\times10^{-2}}\left(1nC + 2nC\right)$

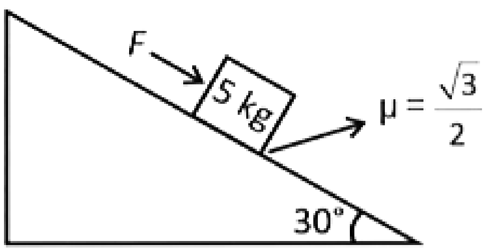
$V_c = \frac{K(3n)}{3\times10^{-2}} = K\times10^{-7}$

$W = 3\times10^{-9-7}\times 9\times10^{9}$

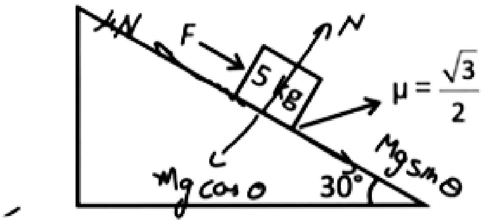
$= 27\times10^{-7}$

$= 2.7\,\mu J$

Question: A block of mass 5 kg is placed on wedge of inclination 30°. Find force applied to move the block downwards with constant speed.



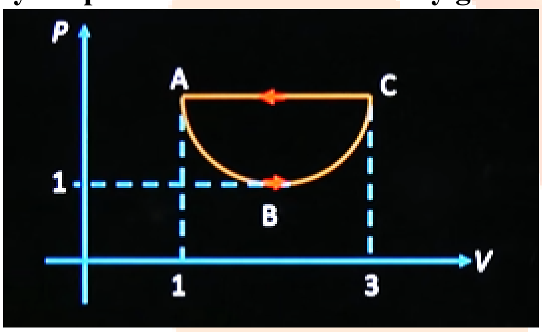
Options:
(a) $\left(\sqrt{3}-1\right)\frac{25}{2}$
(b) 12.5 N
(c) Zero
(d) 25 N
Answer: (b)
Solution:



$$f = \frac{\sqrt{3}}{2} Mg \frac{\sqrt{3}}{2} - \frac{Mg}{2}$$

$$f = Mg\left(\frac{3}{4} - \frac{1}{2}\right) = \frac{Mg}{4}$$

Question: Process ABC represents a parabolic section given by $(V - 2)^2 = 4(P - 1)$ in given cyclic process then work done by gas in process is



Options:

(a) -1/3 units

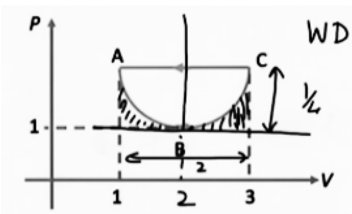
(b) -1/6 units

(c) -1/2 units

(d) -2/3 units

Answer: (a)

Solution:



WD = $-$ive (Enclosed area)

for $x = 1$

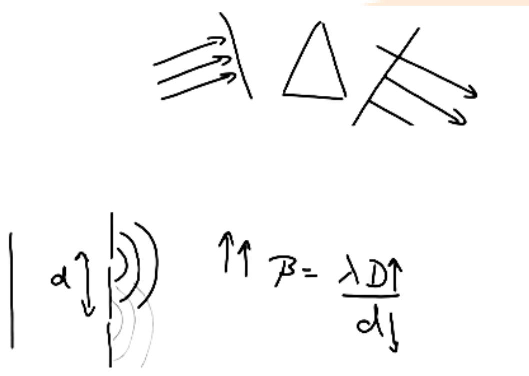
$y = \frac{1}{4}$

$x^2 = 4y$ ← P

Shaded Area $= 2 \times \int_0^1 y\,dx = \frac{1}{2}\,\frac{1}{3} = \frac{1}{6}$ J

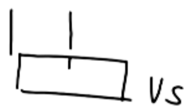
$$WD = -\left(\frac{1}{2} - \frac{1}{6}\right) = -\frac{1}{3}\,J$$

Question: Statement-I: When a planar wavefront passes through a prism then its wavefront doesn't change, but when planar wavefront passes through a smaller slit wavefront becomes cylindrical.
Statement-II: If distance between slits is decreased and screen distance is increased then fringe width increases.
Options:
(a) S-I & S-II are both correct
(b) S-I & S-II are both incorrect
(c) S-I is correct & S-II is incorrect
(d) S-I is incorrect & S-II is correct
Answer: (a)
Solution:



Question: In a vernier callipers when nothing is placed between the jaws zero of vernier scale is ahead of zero of main scale and 4th division consider with one of the main scale. Now when a thin cylindrical wire is kept in the gaps then main scale reading is 15 and 5th vernier division matches with one of the main scale marking. Find the diameter of wire.
(Main scale marking = 1 mm & LC = 0.1mm)
Options:
(a) 15.9 mm
(b) 14.9 mm
(c) 15.8 mm
(d) 15.1 mm
Answer: (d)
Solution:

+ive zero error



$ZE = +0.4\ mm$

$R = 15\ mm + 0.1 \times 5$

$= 15.5\ mm$

$TV = 15.5 - 0.4 = 15.1\ mm$

Question: Two identical cells with same emf ∑ and internal resistance r respectively are given. When cells are connected in series and when they are in parallel in both cases they drive equal current I in external resistance of 6 Ω. Find the value of internal resistance r.

Options:

(a) 3 Ω

(b) 2 Ω

(c) 6 Ω

(d) 7 Ω

Answer: (c)

Solution:

$$\frac{2E}{6+2r} = \frac{E}{\frac{r}{2}+6} = \frac{2E}{r+12}$$

$$6+2r = r+12$$

$$r = \underline{\underline{6\Omega}}$$

Question: In a potentiometer, when a battery is connected with ext. resistance $R_1 = 4\ \Omega$, the balancing length is found to be 120 cm. Now when R_1 is removed and another ext. resistance $R_2 = 12\ \Omega$ is connected then the balancing length is found to be 180 cm. Find internal resistance (in Ω) of the battery.

Options:

(a) 4

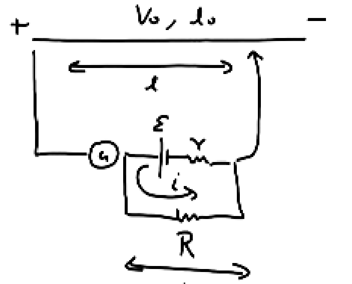
(b) 3

(c) 2

(d) 1

Answer: (a)

Solution:



$$V = iR$$

$$\boxed{\frac{V_0}{l_0} l = V = \frac{E}{r+R} R}$$

$$\frac{V_0}{l_0} 120 = \frac{E \times 4}{r+4}$$

$$\frac{V_0}{l_0} 180 = \frac{E\, 12}{r+12}$$

$$\frac{2}{\cancel{3}} = \frac{1}{\cancel{3}} \frac{(r+12)}{(r+4)}$$

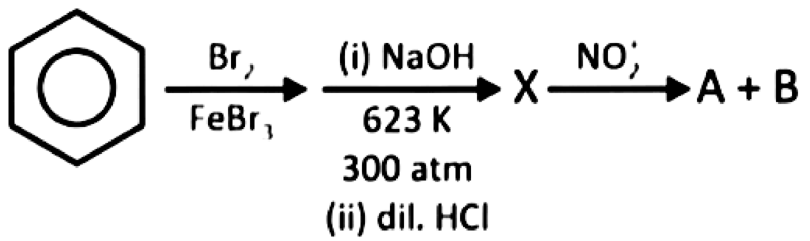
$$2r+8 = r+12 \Rightarrow \underline{r=4}$$

JEE-Main-28-01-2026 (Memory Based)
[MORNING SHIFT]
Chemistry

Question: In Carius method of estimation of 'Br', 1.53 g of an organic compound gave 1 g AgBr. The % of Br in organic compound is, (Atomic mass of Ag, Br = 108, 80 u respectively)
Options:
(a) 35.23
(b) 43.53
(c) 27.81
(d) 22.71
Answer: (c)

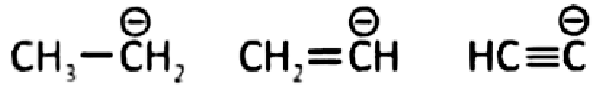
Question: In period 4 of the periodic table which elements have the highest and lowest atomic radii respectively
Options:
(a) K and Br
(b) Na and Cl
(c) K and Se
(d) Rb and Br
Answer: (a)

Question: Consider the following reaction sequence:



The organic product 'A' and 'B' can be separated by
Options:
(a) Steam distillation
(b) Fractional distillation
(c) Distillation under reduced pressure
(d) Azeotropic distillation
Answer: (a)

Question: Consider following ions



(I) (II) (III)

Stability of ions is in order
Options:
(a) III > II > I
(b) II > III > I
(c) I > II > III
(d) I > III > II
Answer: (a)

Question: For a first order reaction, X → Y + Z, time required for decomposition of $\frac{1}{8}th$ and $\frac{1}{10}th$ of its initial conc. Is $t_{1/8}$ and $t_{1/10}$

The value of $\left(\frac{t_{\frac{1}{8}}}{t_{10}}\right)\times 10 =$

Take : log8 = 0.90, log7 = 0.84, log9 = 0.95
Options:
(a) 9
(b) 10
(c) 12
(d) 8
Answer: (c)

Question: For equivalence point X ml of 0.02 M HCl is treated with 5 mL of 0.02 M of a weak base. The pK_b of weak base is 5.69 and the pH of the resulting solution is Y at half of the equivalence point. The value of (x + y) is:
Options:
(a) 5
(b) 8.81
(c) 13.31
(d) 3.81
Answer: (c)

Question: Choose the correct statements in respect of hydrides of Group-15.
A. Reducing power increasing down the group
B. Basic nature increases down the group
C. Stability decreases down the group
D. Boiling point decreases regularly down the group.
Options:
(a) A, B and C only
(b) A, B and D only
(c) A and C only
(d) B, C and D only
Answer: (c)

Question: Which is correct option.
A) $[Ni(CN)_4]^{2-}$ is paramagnetic while $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ are diamagnetic

B) $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$ are diamagnetic while $[NiCl_4]^{2-}$ is paramagnetic
C) $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ are paramagnetic while $[Ni(CN)_4]^{2-}$ is diamagnetic
D) $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ are paramagnetic while $[Ni(CO)_4]$ is diamagnetic
Options:
(a) A
(b) B
(c) C
(d) D
Answer: (b)

Question: The wave number of three spectral lines of H-atom are given. Identify the correct set of spectral lines belonging to Balmer series
Options:
(a) $\frac{5R}{36}, \frac{3R}{16}, \frac{21R}{100}$
(b) $\frac{3R}{4}, \frac{3R}{16}, \frac{7R}{144}$
(c) $\frac{7R}{144}, \frac{3R}{16}, \frac{16R}{255}$
(d) $\frac{5R}{36}, \frac{3R}{16}, \frac{21R}{24}$
Answer: (a)

Question: Given below are two statements
Statement-I: Among XeF_4, BF_4^- and SF_4 the species having equal M-X bond lengths are XeF_4 and BF_4^-. (M = central atom).
Statement-II: Among O_2^{2-}, O_2^-, F_2 and O_2^+ the highest bond order is for F_2 and O_2^{2-}
In the light of the above statements, choose the most appropriate option.
Options:
(a) Both statement-I and Statement-II are correct
(b) Both Statement-I and Statement-II are incorrect
(c) Statement-I is correct but Statemnt-II is incorrect
(d) Statement-I is incorrect but Statemnt-II is correct
Answer: (c)

Question: Among the following-coloured ion is/are
Options:
(a) Ti^{3+} and V^{3+}
(b) Ti^{3+} and Sc^{3+}
(c) Ti^{4+} and V^{3+}
(d) V^{2+} and Sc^{3+}
Answer: (a)

Question: At T(K), 2 moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of ideal solution SO formed is 320 mm Hg. At this stage one mole of A are mixed further, the vapour pressure is found to be 340 mm H. The vapour pressure of pure A and B are respectively.
Options:
(a) 200 mm Hg

400 mm Hg
(b) 440 mm Hg
240 mm Hg
(c) 300 mm Hg
400 mm Hg
(d) 240 mm Hg
440 mm Hg
Answer: (b)

Question: Observe the following reaction:

$$C_3H_6Cl_2 \xrightarrow[NaNH_2]{excess} (X) \xrightarrow[H_2O]{Hg^{2+},H^+} (Y)$$

The product (Y) gives which of the following test?

Options:

(a) Tollen's test
(b) Lucas test
(c) Iodoform test
(d) Fehling's test

Answer: (c)

Question: $X \xrightarrow[H_3O^+]{(i)\ CO_2,\ 500\ atm} Y$

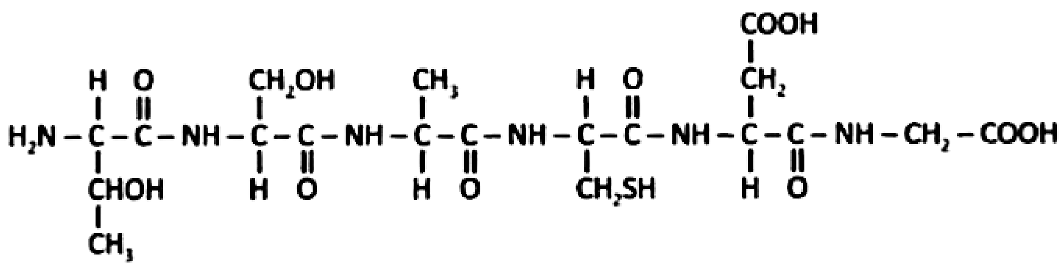
X react with $FeCl_3$
X contain C = 76.57% H = 6.43% O = 17%
V.D. of X = 47
Incorrect statement among following

Options:

(a) X reacts with $NaHCO_3$
(b) X is more acidic than Y
(c) Y is salicylic acid
(d) Y is product of Kolbe's reaction

Answer: (b)

Question: Consider the following polypeptide:



In the given polypeptide, Y is the essential amino acid present. The correct representation of Y and the name of amino acids in the correct sequence in polypeptide is

Options:

(a)

Y	Polypeptide (name of amino acid)
Thr	Thr-Ser-Ala-Cys-Asp-Gly

(b)

Y	Polypeptide (name of amino acid)
Ser	Ser-Ala-Thr-Cys-Asp-Gly

(c)

Y	Polypeptide (name of amino acid)
Thr	Thr-Ser-Cys-Asp-Ala-Gly

(d)

Y	Polypeptide (name of amino acid)
Set	Thr-Ser-Ala-Asp-Cys-Gly

Answer: (a)

Question: Which of the following reaction is correctly matched with the product formed?

Options:

(a) $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2 + Br_2 + KOH \longrightarrow CH_3NH_2$

(b) $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2 + Br_2 + KOH \longrightarrow CH_3-CH_2-NH_2$

(c) $CH_3-CN \xrightarrow{LiAlH_4} CH_3-NH_2$

(d) $CH_3-NC \xrightarrow{LiAlH_4} CH_3-CH_2-NH_2$

Answer: (a)

Question: Match the column-I showing compounds with column-II showing suitable test for that compound

Column-I	Colum-II
P) $C_6H_5COCH_2CH_3$	a) Iodoform test
Q) C_6H_5CHO	b) 2, 4-DNP test
R) $C_6H_5CH_2CHO$	c) Tollen test
S) $C_6H_5COCH_3$	d) Fehling test

Options:

(a) P — b
Q — b, c
R — b, c, d
S — a, b

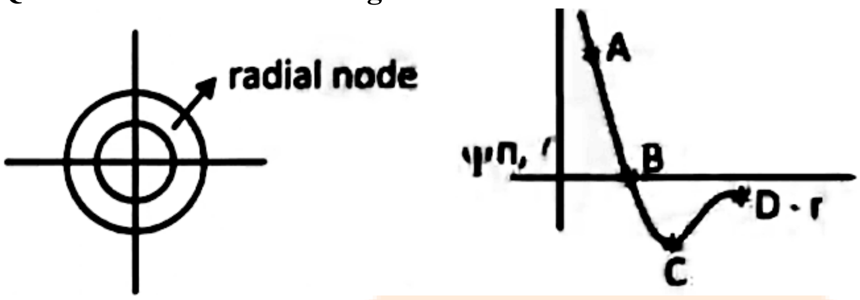
(b) P — b
Q — b, c, d
R — b, c, d
S — a, b, c

(c) P — a, b
Q — b, c, d
R — b, c, d
S — a, b, d

(d) P — b

Q — b, c, d
R — b, c
S — a, b
Answer: (a)

Question: Consider the diagram



Radical node is shown by
Options:
(a) A
(b) B
(c) C
(d) D
Answer: (b)

Question: In reversible isothermal process at 600 K, pressure changes from 0.5 MPa to 0.2 MPa, then
and q. Given moles of gas in container is 1 mol. (R = 8.3 JK^{-1} mol^{-1})
Options:
(a) $\Delta U = 0$
q = -4.587 kJ
w = +4.587 kJ
(b) $\Delta U = 0$
q = 0
w = 0
(c) $\Delta U = 0$
q = 0
w = -4.587 kJ
(d) $\Delta U = 0$
q = +4.587 kJ
w = -4.587
Answer: (d)

Question: Calculate the sum of number of geometrical isomers of [MCl BrNO$_2$CN], number of optically inactive isomers of [M(OX)$_2$Cl$_2$] and number of geometrical isomers of [MCl$_3$Br$_3$]
Options:
Answer: (15)

Question: Consider a galvanic cell reaction:

$$3BH_4^-(aq) + 4ClO_3^-(aq) \xrightarrow{H^+} 4Cl^- + 3H_2BO_3^-(aq) + 3H_2O$$

EMF of cell is given by,

$E_{cell} = E^{o}_{cell} - \frac{RT}{nF} ln[Q]$

Here ‘Q’ is reaction quotient for the given cell reaction. Find ‘n’.

Options:

Answer: (24)

Question: 500 ml of 1.2 M KI solution is reacting with 500 ml of 0.2 M $KMnO_4$ solution, and product iodine is further reacting with 0.1 M $Na_2S_2O_3$ solution. The valume of $Na_2S_2O_3$ solution required for complete reaction is_____ ml.

Options:

Answer: (5000)

JEE-Main-28-01-2026 (Memory Based)
[MORNING SHIFT]
Maths

Question: Consider the 10 observations 2, 3, 5, 10, 11, 13, 15, 21, a and b such that mean of observation in 9 and variance is 34.2. Then the mean deviation about median is
Options:
(a) 3
(b) 5
(c) 6
(d) 7
Answer: (a)

$$k = \tan^{-1}\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}\frac{2}{3}\right) + \tan^{-1}\left(\frac{1}{2}\left(\frac{\pi}{2} - \cos^{-1}\frac{2}{3}\right)\right)$$
$$= \tan^{-1}\left(\frac{\pi}{4} + \theta\right) + \tan^{-1}\left(\frac{\pi}{4} - \theta\right)$$
$$= \frac{1+\tan\theta}{1-\tan\theta} + \frac{1-\tan\theta}{1+\tan\theta}$$
$$= \frac{2\left(1+\tan^2\theta\right)}{1-\tan^2\theta} = \frac{2}{\cos 2\theta} = \frac{2}{\frac{2}{3}} = 3$$
$$\sin^{-1}(3x-1) = \sin^{-1}x - \cos^{-1}x$$
$$\sin^{-1}(3x-1) = \sin^{-1}x - \left(\frac{\pi}{2} - \sin^{-1}x\right)$$
$$\sin^{-1}(3x-1) = 2\sin^{-1}x - \frac{\pi}{2}$$
$$\Rightarrow x = 0$$
$$\text{Only 1 solution}$$

Question: If α, β are roots of quadratic equation $\lambda x^2 - (\lambda + 3)x + 3 = 0$ and $\alpha < \beta$ such that $\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{3}$, then find sum of all possible values of λ.
Options:

(a) 3
(b) 2
(c) 4
(d) 6
Answer: ()

$$\frac{\beta-\alpha}{\alpha\beta}=\frac{1}{3}$$

$$\Rightarrow g(k-\alpha)^2=(\alpha\beta)^2$$

$$\Rightarrow g\left((\alpha+\beta)^2-4\alpha\beta\right)=(\alpha\beta)^2$$

$$g\left(\left(\frac{n+3}{n}-\frac{4.3}{n}\right)\right)=\frac{9}{n^2}$$

$$\Rightarrow d^2-6d+8=0$$

$$\alpha+\beta=\frac{N+3}{N}m,\alpha\beta=\frac{3}{n}$$

Question: If 3 balls are taken from the box without replacement and found to be all black. If all configuration of red balls and black balls are equally likely then the probability that box contained 1 red and 9 block ball is $\frac{p}{q}$ for some coprime natural number p and q then, p + q is
Options:
(a) 59
(b) 69
(c) 57
(d) 79
Answer: ()

$$p\left(\frac{IR}{BBB}\right)=\frac{p(1R\cap BBB)}{P(BBB)}$$

$$P(BBB)=\frac{1}{11}\left(0+0+0+\frac{{}^3C_3}{{}^{10}C_3}+\frac{{}^4C_3}{{}^{10}C_3}+\frac{{}^5C_3}{{}^{10}C_3}\ldots\ldots\frac{{}^{10}C_3}{{}^{10}C_3}\right)$$

$$p(1R\cap BBB)\Rightarrow\frac{1}{11}\left(\frac{{}^9C_3}{{}^{10}C_3}\right)=\frac{{}^9C_3}{{}^{11}C_4}=\frac{14}{55}$$

Question: Find the value of $\sum_{k=1}^{\infty}\frac{(-1)^{k+1}\cdot k(k+1)}{k!}$
Options:

(a) $\dfrac{2}{e}$
(b) $\dfrac{3}{e}$
(c) $\dfrac{1}{e}$
(d) e
Answer: (c)

$$\sum(-1)^{k+1}\frac{(k+1)}{(k-1)} \Rightarrow \sum(-1)^{k+1}\frac{(k-1+2)}{(k-1)!}$$

$$=\sum_{k=1}^{\infty}(-1)^{k+1}\left(\frac{1}{(k-1)}+2\left(\frac{1}{(k-1)!}\right)\right)$$

$$\Rightarrow\left(-\tfrac{1}{0!}+\tfrac{1}{1!}-\tfrac{1}{2!}+\tfrac{1}{3!}\right)+2\left(1-\tfrac{1}{1!}+\tfrac{1}{2!}+\tfrac{1}{3!}+\ldots..\right)$$

$$\Rightarrow-\left(1-\tfrac{1}{1!}+\tfrac{1}{2!}\ldots\ldots.\right)+2\left(1-\tfrac{1}{1!}+\tfrac{1}{2!}\ldots\ldots.\right)$$

$$\Rightarrow-e^{-1}+2\left(e^{-1}\right)$$

$$=\tfrac{1}{e}$$

Question: $\lim\limits_{x\to 0}\dfrac{\ln\left(\sec(ex)\sec\left(e^{2}x\right)\sec\left(e^{3}x\right)\ldots\sec\left(e^{10}x\right)\right)}{e^{2}-e^{2}\cos x}$
Options:
(a) $\dfrac{e^{18}-1}{e^{2}-1}$
(b) $\dfrac{e^{20}-1}{e^{2}-1}$
(c) $\dfrac{e^{16}-1}{e^{2}-1}$
(d) $\dfrac{e^{22}-1}{e^{2}-1}$
Answer: (a)

$$\lim_{x\to 0}\frac{\ln(\sec(ex)+\ln+(\sec(e^2x))+\ln(\sec(e^3x))+\ldots.\ln(\sec(e^{10}x)))}{e^2(1-\cos x)}$$

$$\Rightarrow \frac{e\tan(ex)+e^2\cdot\tan(e^2x)+\ldots..e^{10}\tan(e^{10}x)}{e^2\left(\frac{\sin x}{1^x}\right)\cdot x}$$

$$= \frac{1}{e^2}\left(\frac{e^2\cdot\tan(ex)}{ex}+e^4\cdot 1+\ldots..e^{20}\right)$$

$$\frac{e^{20}-1}{e^2-1}$$

$$= 1+e^2+e^4+\ldots..e^{18}$$

Question: Consider a circle C_1 passing through origin and lying in region x ≥ 0 only, with diameter 10. Consider a chord PQ of C_1 with equation = y and another circle C_2 which has PQ as diameter. A chord is drawn to C_2 passing through (2, 3) such that distance of chord from centre of C_2 is maximum has equation x + ay + b = 0 then (b - a) is

Options:

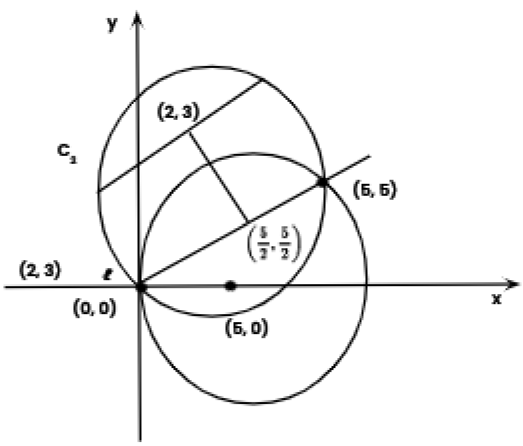
(a) 4

(b) 3

(c) 2

(d) 5

Answer: (c)



$$(x-5)^2+y^2=25$$

$$x-y+c=0$$

$$2-3+c=0$$

$$C=1$$

$$x-y+1=0$$

$$a=-1, b=1$$

$$b-a=2$$

Question: If y = f(x) satisfies the differential equation
$x\frac{dy}{dx}-\sin^2 y=x^3\cos^2 y$ and $y(1)=\frac{\pi}{4}$, then $y\left(\frac{\pi}{3}\right)$ **is**
Options:
(a) 1
(b) $\tan^{-1}\left(\frac{\pi}{4}\right)$
(c) $\tan^{-1}\left(\frac{\pi^3}{27}\right)$
(d) zero
Answer: (c)

$$\Rightarrow \sec^2 y\frac{dy}{dx} - \frac{2\sin y.\cos y}{x\cdot\cos^2 y} = x^2$$

$$\Rightarrow \sec^2 y\cdot\frac{dy}{dx} - \frac{2\tan y}{x} = x^2$$

$$\tan y = t$$

$$\sec^2 y\cdot\frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - \frac{2}{x}t = x^2$$

$$If = \frac{1}{x^2}$$

$$t\frac{1}{x^2} = x + c$$

$$\Rightarrow \tan y = x^3 + c\cdot x^2$$

$$1 = 1 + c \Rightarrow c = 0$$

$$y = \tan^{-1}\left(x^3\right)$$

$$\tan^{-1}\left(\frac{\pi^3}{27}\right)$$

Question: Product of first three term of G.P is 27, then the range of sum of these terms is R-(a, b) then $a^2 + b^2$ is
Answer:

$$\frac{d}{r}, a, ar$$

$$a^3 = 2+ \Rightarrow a = 3$$

$$s = 3\left(\frac{1}{r} + r + 1\right)$$

$$r > 0 \quad s \geq 9$$

$$r < 0 \quad s \leq -3$$

$$s\sum(-0,-3] \cup [9,0)$$

$$R - (-3, 9)$$

$$a^2 + b^2 = 90$$

Question: Let z be a complex number lying in the first quadrant such that $|z - 6| = 5$ and $|z - 3i + 5| = 7$, then $z^3 - 7z^2 + 25z + 16$ is equal to
Options:

(a) 45
(b) 55
(c) 35
(d) 25
Answer: (b)

$$\left.\begin{array}{l}(x-6)^2+y^2=25\\(x+5)^2+(y-3)^2=49\end{array}\right\}\rightarrow(2,3)$$

$$z=2+3i$$

$$(z-2)^2=9$$

$$\Rightarrow z^2+4-4z=-9$$

$$\Rightarrow z^2-4z+13=0$$

$$Z\left(z^2-4z+13\right)-3z^2+12z+16$$

$$=55$$

Question: Consider polynomial functions f(x) and g(x) such that g(x) = $3x^2$ + 2x - 3, f(0) = -3 such that 4g(f(x) = $3x^2$ - 32x + 72 then f(g(2)) is equal to
Options:
(a) $-\frac{25}{6}$
(b) $\frac{25}{6}$
(c) $-\frac{7}{2}$
(d) $\frac{7}{2}$
Answer: (d)

$$g(x) = 3x^2 + 2x - 3$$

$$g(f(x)) = \frac{1}{4} \cdot 3x^2 - 8x + 18$$

$$= 3\left(\frac{x}{2}\right)^2 - 8x + 18$$

$$f(x) \rightarrow \text{linea function}$$

$$f(x) = \frac{1}{2}x + c$$

$$= \frac{x-6}{2}$$

$$g(2) = 13$$

$$f(13) = \frac{7}{2}$$

Question: Let three unit vectors are $\overrightarrow{a}, \overrightarrow{b} \text{ and } \overrightarrow{c}$ such that $\left|\overrightarrow{a} - \overrightarrow{b}\right|^2 + \left|\overrightarrow{b} - \overrightarrow{c}\right|^2 + \left|\overrightarrow{c} - \overrightarrow{a}\right| = 9$ Then the value of natural number k such that $\left|2\overrightarrow{a} + k\overrightarrow{b} + k\overrightarrow{c}\right|^2 = 9$ is

Options:

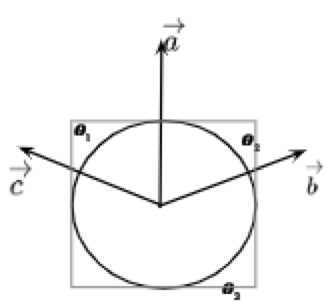
(a) 9

(b) 6

(c) 3

(d) 5

Answer: (d)



$$\sum 2-2\left(\vec{a}\cdot\vec{b}\right)=9$$

$$\Rightarrow 6-2\left(\sum \cos\theta_1\right)=9$$

$$\Rightarrow \sum \cos\theta_1=-\frac{3}{2}$$

$$\Rightarrow \cos\theta_1=\cos\theta_2=\cos\theta=-\tfrac{1}{2}$$

$$4+k^2+k^2+2\left(2k\left(-\tfrac{1}{2}\right)+k^2\left(-\tfrac{1}{2}\right)+2k\left(-\tfrac{1}{2}\right)\right)=9$$

$$(k-2)^2=9$$

$$\Rightarrow k-2=3$$

$$k=5$$

Question: The common difference of AP a_1, a_2, a_3 …….a_m is 13 times the common difference of Ap b_1, b_2, b_3 ….. B_n. Also a_{78} = 327, b_{43} = - 385, b_{31} = -277, then a_1 is equal to

Answer:

$$\mathrm{d}_1=13(d_2)$$

$$a_1+77d_1=327$$

$$b_1+42d_2=-385$$

$$b_1+30d_2=-277$$

$$b_1=-7, d_2=-9$$

$$d_1=13(-9)$$

$$a_1=9336$$

Question: Let $A=\begin{bmatrix}12 & -5\\ 10 & 6\end{bmatrix}, B=\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}12\\ -6\end{bmatrix}$ **such that**

B = (I + A)$^{-}$1, then x_1 - x_2 is equal to

Options:

(a) 27

(b) 108

(c) 21

(d) 54

Answer: (b)

$$\Rightarrow (1+A)^{-1}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}[1+A]$$

$$= \begin{bmatrix} 12 \\ -6 \end{bmatrix}\begin{bmatrix} 13 & -5 \\ 10 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 186 \\ 78 \end{bmatrix}$$

$$x_1 = 186$$

$$x_2 = 78$$

$$x_1 - x_2 = 108$$

Question: Let a line $L : x + 2\sqrt{2}y - 4 = 0$ **cuts x-axis and y-axis at A and B respectively. Consider an equilateral triangle ABC, such that (0, 0) is the orthocentre of ΔABC. If C**$\in(\alpha, \beta)$ **then** **is**

Options:

(a) 4

(b) 6

(c) 2

(d) 3

Answer: (b)

$$\frac{\alpha+4+0}{3} = 0$$

$$\Rightarrow \alpha = -4$$

$$\frac{0+\sqrt{2}+\beta}{3} = 0$$

$$\Rightarrow \beta = -\sqrt{2}$$

$$\left(-4, -\sqrt{2}\right)$$

$$\left|\alpha + \sqrt{2}Gb\right| = 6$$